

National University of Singapore
School of Computing

Semester 1, AY2026-27

CS4246/CS5446

Reinforcement Learning and Sequential Decision Making

Issued: 19 Aug 2026

Tutorial 1 Week 3: Real World Planning and Acting

Guidelines

You may discuss the content of the questions with your classmates. But everyone should work on and be ready to present ALL the solutions.

Instructions for Tutorial Team Sign-up

- All students must be part of a Tutorial Team (2–4 members).
- Teams are created and managed through Canvas.
- To join or create a team:
 1. Log in to Canvas and navigate to the course site.
 2. Go to the menu item *People* → *Groups* → *Tutorial Teams*.
 3. Browse the available groups for your tutorial section.
 4. Either:
 - join an existing group with available slots, or
 - form a new group and invite classmates.
 5. The members of your group may be enrolled in *different* tutorial sessions.
- Deadline: teams should be finalised on Canvas by the end of Week 2.

Problem 1 – A Blocks World

Consider a blocks-world as shown in the figure below. The objective of this problem, called *Middle-pop*, is to remove the block B in-between two other blocks, A and C, and put it on the Table (represented by the horizontal line).



- a. Express the blocks-world problem as a planning problem in the Planning Domain Definition Language (PDDL). You may use the following assumptions, and/or make **additional** assumptions as necessary:
- There are only four objects in the world – Block A, Block B, Block C, and the Table.
 - A set of predicates define the descriptions of and the relations among the objects:
 - A block can be on top of another object, i.e., **On**(x, y), where x is a block and y is another object.
 - The top of the object to be clear, i.e., **Clear**(x), where x is an object, before another object can be placed on top of it.
 - You can define additional predicates if necessary.
 - A set of action schema and actions define how to change the world state:
 - A block can be re-arranged by moving it to (the top of) another object, i.e., **Move**(x, fr, to), where x is a block, and fr, to are other objects in the world.
 - A block can be moved only when it is not stacked upon, i.e., it is “clear” on the top, or **Clear**(x), where x is a block.
 - You cannot move the Table, but you can always move something to or from the Table, i.e., **Clear**(Table) is always assumed to be True (interpreted as: there is always “clear space” on the Table).
 - You can define additional action schemas or actions as necessary.
- b. Show the search tree for planning by forward search.
- c. Show a valid plan for the problem.

Problem 2 – Heuristic for Planning

In classical planning, suppose goals and action preconditions are restricted to positive literals. Explain why removing all negative effects from every action schema results in a relaxed problem. Does this still hold if negative preconditions or goals are allowed?

Problem 3 – Hierarchical Task Planning

Context We consider a hierarchical task planning problem where the world is described by five fluents: A, B, C, D, E . Each fluent can be true (positive) or false (negative).

The following primitive actions are available:

- Action($P1$, Precond: $\neg A$, Effect: A)

- Action(P2, Precond: A , Effect: B)
- Action(P3, Precond: A , Effect: $\neg B$)
- Action(P4, Precond: $\neg B$, Effect: $A \wedge C$)
- Action(P5, Precond: $\neg B$, Effect: C)
- Action(P6, Precond: $\neg B$, Effect: $A \wedge \neg C$)
- Action(P7, Precond: $\neg B$, Effect: $\neg C$)
- Action(P8, Precond: $\neg B \wedge \neg C$, Effect: D)
- Action(P9, Precond: D , Effect: E)
- Action(P10, Precond: $\neg B \wedge D$, Effect: E)

Notation

- $\wedge$ means logical conjunction.
- $\neg$ means logical negation.
- $\sim +$ denotes “possibly add”, $\sim -$ denotes “possibly delete”, $\sim \pm$ denotes “possibly add or delete”.
- $[P1, P2, \dots]$ denotes a sequence of primitive actions (or an HLA implementation).

A) Define schemas for the following higher-level actions (HLAs), with their preconditions, effects, and possible implementations.

i. Action(H1, Precond: $\neg A$, Effect: _____)

Possible implementations:

- [P1, P2]
- [P1, P3]

ii. Action(H2, Precond: _____, Effect: _____)

Possible implementations:

- [P4]
- [P5]
- [P6]
- [P7]

iii. Action(H3, Precond: _____, Effect: $D \wedge E$)

Possible implementations:

- [P8, P9]
- [P8, P10]

B) Consider the planning problem:

- Initial state: $\neg A \wedge B$
- Goal state: E

Assuming negative literals are allowed in the initial state (closed world assumption), determine which sequences of HLAs with their implementations achieve the goal. Choose all that apply.

- i. H1 (P1,P3), H2 (P6), H3 (P8,P9)
 - ii. H1 (P1,P3), H2 (P7), H3 (P8,P10)
 - iii. H1 (P1,P2), H2 (P4), H3 (P8,P9)
 - iv. H1 (P1,P2), H2 (P5), H3 (P8,P9)
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